

S10 — adversarial review, with responses

2026-08-02. Twelve findings from the adversarial methodology review, each with the attack stated plainly, our response, and what has been done or is queued. A glossary of every term of art is at the end (§4); terms are marked with a dagger† on first use.

Three findings were re-derived independently from the raw data before being accepted, because a review that reports 48 of 49 findings surviving its own verification is itself a thing to be suspicious of. All three reproduced exactly. The rest are accepted on the review's arithmetic where that arithmetic is checkable against files we hold.

1. Summary

#	Finding	Severity	Our response	Status
F1	The word-level mechanism test is a rarity index, not an AI-preference score	Fatal	Accept — verified independently	Withdrawn; replacement built
F3	Kobak's content words — the field's standard negative control† — were never run, and v1.0 fails them	Major	Accept; the fix strengthens the paper	Done (v1.1)
F2	Placebo pool† matched on frequency but not dispersion†	Major	Accept	Done (v1.1)
F5	Sensitivity† measured only on unedited text — the class we argue is absent	Major	Accept — biggest framing change	Restating; new Se tier queued
F8	One ratio pooled over 2024–26 is not a stable estimand†	Major	Accept — and it dents a claim I made	Per-year series queued
F4	The NB prevalence figure is a five-month window, and §5 names the wrong stratifier†	Major	Accept	Restating
F6	The frequency-weighted secondary does not replicate; the sweep claim is misquoted	Major	Accept	Correcting; full table queued
F7	The formality control is not actually frequency-matched	Major	Accept	Rebuild queued
F9	Zero-count smoothing† is asymmetric between instrument and placebo	Minor	Accept	Subsumed by F3
F11	Instrument citation wrong (fourth author is Lause, not Berens)	Minor	Accept	Fixed
F12	No-interference† assumption never stated	Minor	Accept	One paragraph queued
F10	Estimand departs from Kobak's, forfeiting the lower-bound reading	Minor	Partly accept	Parallel estimand queued

Net effect on the headline. Effect sizes roughly halve. The combined p-value goes from 2.4×10^{-7} to 5.7×10^{-5} . Two of five confirmatory chambers become individually null. The claimed prevalence becomes a lower bound over a five-month window rather than a corpus rate. **Nothing reverses direction.**

2. Findings, and what we are doing

F1 — Fatal. The mechanism test measured rarity, not AI preference

The attack. §6's third and strongest anti-p-hacking† defence claimed that a word's "AI preference", measured only from synthetic AI speech and a 2019 control, predicts which words later rose in Hansard — Spearman† +0.548, permutation $p^\dagger < 10^{-4}$. But the 2019 control is a *subset* of the 2018–22 pre-window, so each word's rarity sits inside both the predictor and the outcome with the same sign.

Independently verified. Every number reproduced:

quantity	Spearman
ai_pref vs shift (the published claim)	+0.548
ai_pref vs $-\log(2019 \text{ control count})$	+0.923
ai_pref vs the actual synthetic-corpus count	-0.234
partial correlation† controlling $\log(\text{pre count})$	-0.008

The original permutation shuffled globally, which destroys the frequency structure and therefore cannot see the confound†. Shuffling *within* frequency strata† gives **p = 0.24** (0.32 at ten strata). The predictor correlates *negatively* with the synthetic evidence it was supposed to encode.

Response: accept in full. The claim is withdrawn in `METHODOLOGY.md`. Reporting +0.548 next to a partial of -0.008 would be an admission, not a defence.

Replacement, already running. The fix is a **paired†** design, as in Geng & Trotta and Yakura et al.: compare an instruction-tuned model against *its own base model* on identical prompts, so each word's corpus frequency cancels exactly. First results on two model families (Qwen3-8B, Mistral-7B):

- Kobak style words are strongly preferred by instruct over base: excess **+0.88** over frequency-matched controls, $p < 0.001$ — i.e. the "AI register" is substantially the *alignment* register
- that paired score predicts the Hansard shift: Spearman +0.148, partial +0.134, **stratified permutation p = 0.031**
- the unpaired rival (generic model-text vs human text) predicts nothing: Spearman -0.000, $p = 0.79$
- rarity check: the new predictor scores +0.23 against log-rarity, where the dead one scored +0.92

F3 — Major. The standard negative control was never run

The attack. Kobak's team annotated all 900 excess words content-versus-style *blinded to year*, precisely so the two could be separated. The 462 content words are biomedical topic vocabulary and are the field's canonical topic-churn control. No script in the study touched them. Run under v1.0, the primary statistic **cannot tell them apart from the style list** — Canada +0.282 content versus +0.210 style.

Response: accept, and it makes the paper stronger. The cause is benign: content words are largely absent from Hansard, and each absent word collects a constant positive term from the smoothing (F9). Restricting both instrument and placebo pool to words **present in both periods** fixes it. Now run in all six chambers:

chamber	style excess	content excess	style – content
New Brunswick	+0.278	−0.004	+0.282
UK Commons	+0.121	−0.258	+0.379
Dáil Éireann	+0.080	−0.354	+0.434
US House	+0.059	−0.167	+0.226
Canada Commons	+0.029	−0.450	+0.479
US Senate	−0.026	−0.083	+0.057

Content words go sharply negative everywhere. The instrument separates cleanly from a topic list drawn by the same authors from the same source by the same method. **This is a clean specificity† result and it was sitting unpublished in our own data directory.**

F2 — Major. Placebos matched on frequency but not dispersion

The attack. Placebo words were matched to instrument words on pre-period frequency only. Corpus linguistics treats **dispersion†** — how many distinct sittings a word appears in — as co-equal with frequency (Gries 2008; Egbert & Biber 2019). A pool full of bursty topic words decays faster between windows, depressing the placebo median and inflating every excess. We had already written this exact filter for the formality control after it returned place names, and never ported it.

Response: accept. v1.1 matches on both axes. Costs 12–44% of each effect size. All chambers that were significant remain so except Canada.

F5 — Major. Sensitivity was measured on the wrong class of text

The attack. The sensitivity† figure ($Se = 40/40$) comes from 40 synthetic speeches, one generator, one sampling configuration, **unedited**. But §4.1 concludes the record contains very little unedited paste, and §5.1 says the signature is *edited* AI. So the detector's measured sensitivity is on precisely the class of text we argue is absent. Published work confirms the gap: Liang et al. ran the equivalent proofreading arm and found lightly-edited text yields only a slight, non-significant rise; Dugan et al. show detector performance collapsing under light edits across eight detectors.

Sensitivity analysis: at $Se = 0.5$ the prevalence estimate goes $7.5\% \rightarrow 15.0\%$; at $Se = 0.3$, $\rightarrow 25.0\%$.

Response: accept. This is the single most important change to what we may claim. "The Rogan–Gladen† correction is the identity" is deleted. The prevalence figure becomes **a lower bound at Pangram's operating point, conditional on unmeasured sensitivity to edited AI** — defensible floor, no upper bound. Interval widens to roughly [3.0%, 14.8%] once Se/Sp sampling error is propagated.

Queued. Generate a light-edit tier of the same 40 speeches and report Se separately for raw, edited, and outline-expanded text.

F8 — Major. Pooling 2024–2026 into one ratio, and what it does to Canada

The attack. The marker vocabulary is known to be non-stationary† inside the post window — Geng & Trotta show *delve* collapsing from early 2024 once the press named it, while other markers keep rising — and the chambers weight the post years very differently. Canada is **49% 2024** by words; Ireland 31%. Per-year primaries:

chamber	2024	2025	2026
Canada	−0.036	+0.099	+0.090
Ireland	+0.166	+0.171	+0.179
UK	+0.195	+0.196	+0.161

Response: accept — and this dents something I told you. I explained Canada's weak lexicon result as a national or behavioural difference, and built a story around Canada being the drafting-heavy chamber. A sufficient and more boring explanation is that **half of Canada's post-window is 2024**, its weakest year. Canada's 2025 and 2026 values (+0.099, +0.090) are ordinary. That does not undo the Pangram prevalence result, which is 2025–26 only, but it does mean the lexicon half of the Canada story needs restating.

Queued. Report the primary as a per-year series in every chamber and disclose the post-year mixture in the cross-corpus table.

F4 — Major. The prevalence figure's reference population is wrong

The attack. Two errors in one sentence. The sampling frame is 3,998 segments from five sitting months (Oct/Nov/Dec 2025, Mar/May 2026) — 10.6% of the corpus, omitting all of 2024 and most of 2025, sitting on the steepest part of the ramp. And §5 says the sample was stratified† by "the screen score", but the code ranks by the **Falcon/Binoculars** score — the detector §4.1 reports as having found nothing. The realised verdicts confirm it was flat across deciles (0,1,1,1,1,0,1,1,1,2).

Response: accept. Restated as "7.5% [4.0–13.6] of New Brunswick segments in sittings from October 2025 to May 2026". §5 corrected to say Falcon deciles, with a note that the stratifier was uninformative so the design degenerated to an effective simple random sample — which is harmless for the point estimate under equal allocation, but must be said. The other three chambers use a different window again (from 2025-01-01) and that is now labelled.

F6 — Major. The frequency-weighted secondary does not replicate

The attack. §6 claimed "every cutoff from <1 to <50 per 100k gives $p \leq 0.008$." That is New Brunswick only, unlabelled, and misquoted — NB's <1/100k cell is $p = 0.015$. Across the three confirmatory chambers, **one of eighteen cells clears 0.05**.

Response: accept. Correct the figure, label the chamber, print the sweep for all chambers, and state plainly that the frequency-weighted secondary does not replicate — then make the argument for equal weighting explicitly rather than letting a reader assume the two statistics agree. §2.2 already shows the pooled statistic is null in NB, so the design choice is defended; the sweep sentence was simply wrong.

F7 — Major. The formality control is not frequency-matched

The attack. §3.1 says the control is frequency-matched bucket-for-bucket. The code discards candidates with pre-count < 200, so the buckets holding most of the instrument cannot be filled. Realised control sizes against the instrument's 407: NB 67, Canada 164, Ireland 201, UK 211. The control is therefore systematically *more common* than what it controls. The showcased register vocabulary is also UK-only; Ireland's run returns *dirty*, *terribly*, *toilet*, *bread* and two untagged Irish tokens.

Response: accept. Lower the floor, fill quotas within bucket, print the realised match table, show all chambers' control words. Note the study already reports both baselines — the smaller "vs formality control" numbers (+0.088 UK / +0.052 IE / +0.090 CA) are in the progress report — but the headline picked the more favourable one without reconciling them.

F9 — Minor. Smoothing is asymmetric between instrument and placebo

Because the pre-window is about twice the post-window, a word absent from *both* periods still scores a positive constant (+0.58 to +0.68). The genuinely new part is the placebo side: the placebo pool is built from pre-period counts, so its lowest bucket holds words with count 1, whose expected log fold-change is about -0.43 . The structural gap between an absent instrument word and its matched placebo is therefore ~ 1.1 log units **before any AI exists**. That is the mechanism behind F3. **Subsumed by the present-in-both fix.**

F11 — Minor. The citation is wrong

The fourth author is **Jan Lause**, not Berens — Berens is the lab, which is where the error came from. Web-verified: Kobak D, González-Márquez R, Horvát E-Á, Lause J, *Science Advances* 11(27):eadt3813 (2025), arXiv 2406.07016. **Fixed** in `METHODOLOGY.md` and `replication_protocol.md`. It is also in the project's `CLAUDE.md` and should be corrected there before it reaches the paper.

Scope note also accepted: our 407 style words are the **full 2013–24 annotation**, not the 2024-only excess list. The review is right that the contamination direction is mostly conservative — a word whose break falls inside our pre-window raises the pre baseline and *shrinks* the measured effect.

F12 — Minor. The no-interference assumption is never stated

SUTVA[†] — the assumption that one unit's treatment does not affect another unit's outcome — is nowhere in our documents, even though §5.2 *measures its violation* directly: AI vocabulary rising in text a calibrated detector rates least AI-like. The placebo pool has no semantic exclusion, so placebo words can themselves be contaminated.

Response: accept, and it is free strengthening. Contaminated controls bias the estimate *toward* the null, so every reported effect is conservative under permeation. One paragraph converts an unstated vulnerability into a point in our favour.

F10 — Minor. Partly accepted

Accepted: Kobak's unit is document *containment* — the fraction of abstracts containing at least one group word — and that is what licenses his "13.5% is a lower bound" reading. Our equal-weight mean of per-token log fold-change has no prevalence interpretation, which is exactly why we can claim consistency but not magnitude. Adding containment as a parallel primary (fraction of *turns* containing ≥ 1 style word, observed minus placebo-expected) is cheap and yields a number a reader can act on. Credit Gray for the equal-weight-mean design and Matsui for the control-set idea.

Rejected: "silently changes the estimand" overstates. §2.1 claims Kobak only for the word list, and §2.2 justifies equal weighting on independent grounds without attributing it to anyone. The review concedes both points itself.

3. Attacks that failed — have these answers ready

- **"Canada's significance is placebo contamination."** No. Under like-for-like dispersion matching Canada stays at $p = 0/1000$. The version that weakens it imposes the *instrument's median* dispersion on the placebo pool, making every placebo word more topic-general than half the instrument's own words — the mirror image of the alleged bias.
- **"It can't tell LLM style from topic churn."** Only under the smoothing-contaminated specification. See F3: content words go to $-0.08 \dots -0.59$ while style stays positive.
- **" $p = 2.4 \times 10^{-7}$ is arithmetic, not data."** Partly fair and already conceded — the combination uses bounded p-values, so the exponent tracks the number of placebo draws. But *which* chambers hit the resolution floor is a fact about the data, and NB's real value is $4\times$ the most extreme of 100,000 draws. Fix: lead with the standardised effect size[†], footnote the p-value.

- **"A few verbose speakers drive it."** Covered by the speech-clustered bootstrap†, which resamples whole speeches rather than words.
 - **"The 2013–24 instrument scope inflates the effect."** Wrong direction — it shrinks it (see F11).
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4. Glossary

Written for a reader comfortable with probability and measurement but not with the conventions of empirical social science. Analogies are to instrumentation and signal processing where they help.

Testing and inference

Null hypothesis. The boring explanation you are trying to rule out. Here it is never "nothing changed" — it is "this changed no more than comparable vocabulary changed anyway."

p-value. The probability of seeing a result at least this extreme *if the null were true*. It is not the probability the null is true. Think of it as a false-alarm rate for a detector whose threshold you set at the observed value.

Permutation / placebo test. Instead of assuming a distribution for the null, you *build* it: scramble or re-draw the thing you are measuring many times under conditions where the effect cannot exist, and see where the real value falls in that empirical distribution. Our version draws 1,000 random word sets matched to the instrument and runs the identical statistic on each. This is the empirical-noise-floor approach: measure the noise directly rather than modelling it.

$p < 0.001$ as a bound. With 1,000 draws, the smallest observable p is 1/1000. " $p < 0.001$ " means zero draws beat the real value — the resolution limit of the simulation, not a measurement. More draws would resolve further.

Stratified permutation. A permutation test that shuffles only *within* groups sharing a nuisance property (here, word frequency), so the nuisance structure is preserved and cannot masquerade as signal. A global shuffle destroys that structure, which is exactly how F1's confound went unseen.

Confound. A third variable driving both the thing you measure and the thing you predict, producing a correlation that means nothing. In F1 the confound was the word's own rarity.

Partial correlation. The correlation remaining between two variables once a third has been statistically removed from both — the regression residuals correlated against each other. If the raw correlation is +0.55 and the partial is −0.01, the third variable was doing all the work.

Spearman correlation. Pearson correlation computed on *ranks* rather than raw values. Robust to outliers and to non-linear-but-monotone relationships, which matters for word counts spanning several orders of magnitude.

Fisher's method. Combines independent p-values into one: $X^2 = -2 \cdot \sum \ln(p_i)$, tested against a chi-squared distribution with 2k degrees of freedom. It is how you ask "taken together, is this pattern unlikely?" across several independent studies.

Effect size vs significance. Significance says the effect is unlikely to be noise; effect size says how big it is. With tens of millions of words, everything is significant, so effect size is the only interesting quantity. The **standardised effect size** here is the real value minus the placebo median, expressed in placebo standard deviations.

Power. The probability of detecting an effect that is really there. Small corpora have low power, so a null result from a small corpus is weak evidence of absence — this is why the US Senate null (3M words) and the Canada null (20M words) mean different things.

p-hacking / researcher degrees of freedom. Trying many analyses and reporting the one that worked. The defences are to fix the analysis before seeing the data (pre-registration), to use an instrument someone else built for another purpose, and to have no tunable choices in the primary statistic.

Discovery vs confirmatory corpus. The dataset where you *found* the effect cannot also be evidence for it, because you selected it on the outcome. New Brunswick is our discovery corpus and is reported separately and as an upper bound.

SUTVA / no-interference. The assumption that one unit's treatment does not affect another unit's outcome. Violated here in an interesting way: AI writing changes how *non-AI* text is written, so the controls are partly treated. The direction is conservative — contaminated controls shrink the measured effect.

Measurement and correction

Sensitivity (Se). $P(\text{test flags} \mid \text{the thing is present})$. A detector's true-positive rate. **Specificity (Sp).** $P(\text{test does not flag} \mid \text{absent})$. Its true-negative rate. These are properties of the detector *on your data*, not universal constants, which is why they must be measured in-domain.

Rogan–Gladen correction. An imperfect test's observed positive rate π relates to true prevalence τ by $\pi = \tau \cdot \text{Se} + (1-\tau)(1-\text{Sp})$, so $\hat{\tau} = (\pi + \text{Sp} - 1) / (\text{Se} + \text{Sp} - 1)$. With $\text{Se} = \text{Sp} = 1$ the correction is the identity. The denominator is a variance inflation factor: as the detector worsens, the corrected estimate becomes very noisy.

Clopper–Pearson / Wilson intervals. Confidence intervals for a proportion. Clopper–Pearson is exact and conservative; Wilson behaves better near 0 and 1. Relevant because "40/40" does not mean $\text{Se} = 1.0$ exactly — its 95% lower bound is 0.928.

Rule of three. With zero events in n trials, the 95% upper bound on the rate is about $3/n$. Zero flags in 2,400 pre-LLM segments bounds the false-positive rate at roughly 0.125%.

Two-phase / double sampling. A cheap instrument scores everything and defines strata; an expensive instrument is spent only where informative. The cheap one need not be accurate, only correlated with truth, because the stratum weights are known exactly from the full corpus.

Stratifier. The variable used to divide a population into strata before sampling. If it is uninformative (as ours turned out to be — F4), the design degenerates to simple random sampling, which is not fatal but must be disclosed.

Design-based estimation. Inference that relies only on the known sampling design, not on a model of the data. Robust, and why we sample uniformly at random where we can.

Clustered bootstrap. Resampling whole clusters (here, whole speeches) rather than individual observations, because words within one speech are not independent. Ignoring clustering understates uncertainty, sometimes badly.

Selection on the outcome. Choosing units by the same quantity you then measure in the remainder. Guaranteed to bias the result, because you have removed the upper tail of a noisy distribution. Fixed by **split-half**: rank on one half of the data, measure on the other.

Corpus statistics

Log fold-change. $\log(\text{rate after} / \text{rate before})$. Symmetric around zero and additive, which is why ratios of counts are usually analysed in logs.

Smoothing / pseudocount. Adding a small constant (here 0.5) to counts so zeros do not produce infinities. Not innocent: when the two denominators differ, an absent word gets a systematic non-zero score (F9).

Equal-weight vs pooled (frequency-weighted). Equal-weight gives every word one vote; pooled sums all counts, so common words dominate. They can point opposite ways on the same data — as they do here — so the choice must be declared and defended, not discovered.

Dispersion. How evenly a word is spread across documents or sittings, as opposed to how often it occurs. A word appearing 500 times in one debate and nowhere else is frequent but badly dispersed. Corpus linguistics treats dispersion as co-equal with frequency; ignoring it selects bursty topic words.

Keyness. The corpus-linguistics term for "which words distinguish corpus A from corpus B" — the general problem this study is a special case of.

Frequency matching. Choosing control words with the same baseline frequency as each instrument word, so the comparison is like-for-like. The analogue of matching impedance before comparing two measurements.

Placebo / donor pool. The set from which control instruments are drawn. Its construction determines what the null actually represents; drawing at random from the whole vocabulary is weaker than drawing from a deliberately constructed pool.

Non-stationary. A process whose statistics change over time. Our marker words are non-stationary inside the post-window — some collapsed after 2024, others kept rising — so one pooled ratio over 2.5 years is not a stable quantity (F8).

Paired design. Comparing two conditions on the *same* units, so unit-level nuisance variation cancels exactly. Base-versus-instruct on identical prompts is paired; model-output-versus-human-corpus is not, and that difference is what killed F1 and rescues its replacement.

Burrows Delta. The standard authorship-attribution distance: z-score each document's rate for the most frequent words, then take mean absolute difference. Being a *pairwise* distance, a shift affecting everyone equally cancels out of it — so a rise means genuinely increased heterogeneity.

Shannon entropy / type-token counts. Measures of lexical diversity. Both are strongly biased by sample size, so any comparison must fix the token count per unit first, or it will manufacture the result.

AUC (area under the ROC curve). The probability that a randomly chosen positive scores above a randomly chosen negative. 0.5 is chance, 1.0 perfect. Threshold-free, which is why it is used to compare detectors whose score scales differ.